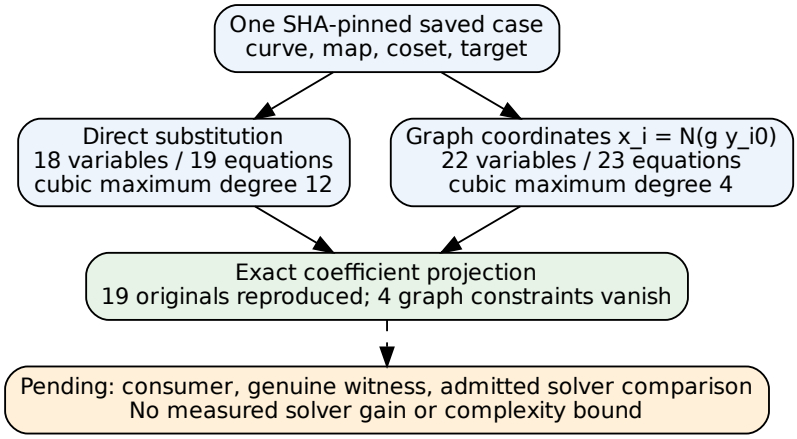
Paired graph builder: exact construction checks

2026-11-13. Unsigned producer evidence [pcc:unverified].

The builder emits direct and graph systems from each identical saved case. Every graph polynomial is symbolically substituted before export: all19 original equations are reproduced coefficient-for-coefficient and all four added graph constraints become zero. Source freeze `00ffe72fbe`; prospective development contract `bf82ba3d4`.

Saved input	Direct $(n, \#eq, \deg_{\max})$	Graph $(n, \#eq, \deg_{\max})$
Reference	$(18, 19, 4)$	$(22, 23, 4)$
Cubic	$(18, 19, 12)$	$(22, 23, 4)$



Controls. All 17 tests pass: four builder controls plus 13 maintained SHA-256 tests. Independent symmetric scalar S3 on two small fields, exact projection for both saved inputs, deterministic lift vectors, changed coefficient/image negative controls and chain boundaries. No genuine elliptic witness is asserted. Original field-audit output and generic Lean inverse-map proof are unchanged; concrete polynomial attachment is checked natively.

Artifacts. JSON coefficients and msolve-style text are pinned by each manifest. External consumer ingestion is unvalidated. The exporter requires one of the two frozen input hashes and creates a fresh output directory. Data statement `PCC-20261113-b52e23` is unsigned, not a signed verdict. Test001 tool status1 conflicts with explicit process receipt0/17 passes; both are retained. Test002 after the output-guard change has tool/process exit0. Release build exit0; five dead-code warnings for development controls.

Interpretation. The previous source directly substituted the cubic; this explicit graph representation was untested. Its reduced input degree justifies a matched comparison. Added variables may erase the benefit; elimination returns the same direct system. No bound on solving degree, critical pairs, memory or runtime follows. The original finite W gate stands.

Next decision. Validate consumer ingestion and a genuine witness; freeze order/resources; obtain live admission; compare complete sparse work on identical inputs. No solver/timing run here. Public-input relation finding and a uniform sub-exponential bit bound remain unestablished.

Sources. Frozen pilot `3466f3be9e`, round5 records/protocol, Galbraith15.8.1–15.8.2, prior auxiliary-cover review. KB unavailable; graph encoding is standard, with no novelty or exhaustive prior-art claim.